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(Approved by AICTE, New Delhi, Affiliated to JNTUH, Hyderabad)
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

II YEAR II SEM SECTION-B **A.Y 2025-26**

RTRP PROJECT ABSTRACT

TITLE OF THE PROJECT: : : SkyRescue AI: Real-Time Disaster Rescue Detection

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ABSTRACT

Natural disasters such as earthquakes, floods, hurricanes, and wildfires often cause widespread destruction, making rapid search-and-rescue operations critical for minimizing casualties and economic loss. Sky Rescue AI presents a YOLO-based real-time detection system designed to identify survivors and assess structural damage using aerial imagery captured by unmanned aerial vehicles (UAVs). Leveraging the high-speed object detection capabilities of the YOLO (You Only Look Once) architecture, the system processes live video streams to detect human presence, injured individuals, fire outbreaks, collapsed buildings, and obstructed pathways with high accuracy and low latency.

The proposed framework integrates optimized deep learning models with edge computing to enable on-device inference, ensuring efficient performance in low-connectivity disaster zones. The dataset comprising annotated aerial images of disaster scenarios used to train and validate the model, enhancing robustness across varying environmental conditions such as smoke, debris, low lighting, and occlusion. Performance evaluation demonstrates high precision and recall rates, along with real-time processing speeds suitable for deployment in time-sensitive rescue missions.

KEY WORDS : Humanitarian technology, Intelligent UAVs, Crisis response innovation, Vision-based survival detection, Edge-powered resilience.

Project Guide : **Project Coordinator :**

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